

Titanic Dataset — Survival Analysis

Data Analysis Project | Beeskilled Data Science with Python Internship

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Program	B.Tech — Computer Science & Engineering
Year	3rd Year
Project	Titanic Dataset — Survival Analysis
Tools	Python, Pandas, Matplotlib, Seaborn

1. Objective

The objective of this project is to analyze the Titanic passenger dataset and identify patterns associated with passenger survival, focusing on gender, passenger class, age, and family size.

2. Dataset and Tools

The dataset contains 891 passenger records and 12 variables, including survival status, passenger class, gender, age, family relationships, fare, cabin, and port of embarkation. Python was used with Pandas, Matplotlib, and Seaborn.

3. Data Preparation

There were 177 missing Age values and 2 missing Embarked values. Missing ages were filled using the median age, and missing Embarked values were filled using the most common value. Cabin was not used in the selected analysis because it contained 687 missing values. No duplicate rows were found.

4. Key Results

Measure	Result
Overall survival	38.38% (342 of 891)
Female survival	74.20%
Male survival	18.89%
1st-class survival	62.96%
2nd-class survival	47.28%
3rd-class survival	24.24%
Small-family survival	57.88%
Large-family survival	16.13%

5. Key Findings

- Only 38.38% of passengers in the dataset survived.
- Female passengers had a survival rate of 74.20%, compared with 18.89% for male passengers.
- First-class passengers had the highest overall class survival rate, while third-class passengers had the lowest.

4. Female passengers had higher survival rates than male passengers within every passenger class. First-class females had a 96.81% survival rate, while third-class males had 13.54%.
5. Survival varied across the age groups used in the analysis. Children had a 57.97% survival rate, while seniors had 26.92%.
6. Passengers travelling with a small family had the highest family-category survival rate at 57.88%; large-family passengers had the lowest at 16.13%.

6. Conclusion

The analysis indicates that survival outcomes on the Titanic were strongly associated with gender and passenger class. Female passengers and passengers in higher classes generally had substantially higher survival rates. Age and family size also showed meaningful differences. These findings provide a concise overview of how passenger characteristics were related to survival in the dataset.

Note: These results describe associations in the dataset and do not establish that any individual factor directly caused survival or death.